Do Now · retrieval

Lesson 121 · Writing Algorithms with Validation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why must a trace table record changes rather than only final output?

Q2

After pushing A then B onto a stack, which is popped first?

Q3

Why must a binary search operate on ordered data?

Q4

An OR gate has inputs 0 and 1. What is the output?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define constraints — State valid range, type and required behaviour before coding.

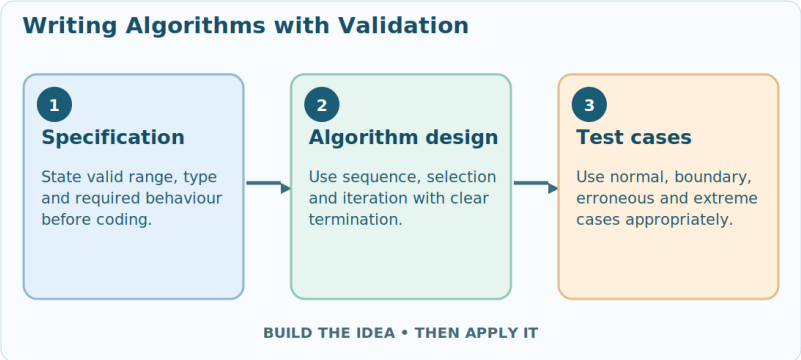
Core idea 2

Build control flow — Use sequence, selection and iteration with clear termination.

Core idea 3

Prove with tests — Use normal, boundary, erroneous and extreme cases appropriately.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

validation — Checking that input meets rules.

robustness — Ability to handle invalid/unexpected input safely.

iteration — Repetition based on loop condition.

boundary — Input at/near permitted limit.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

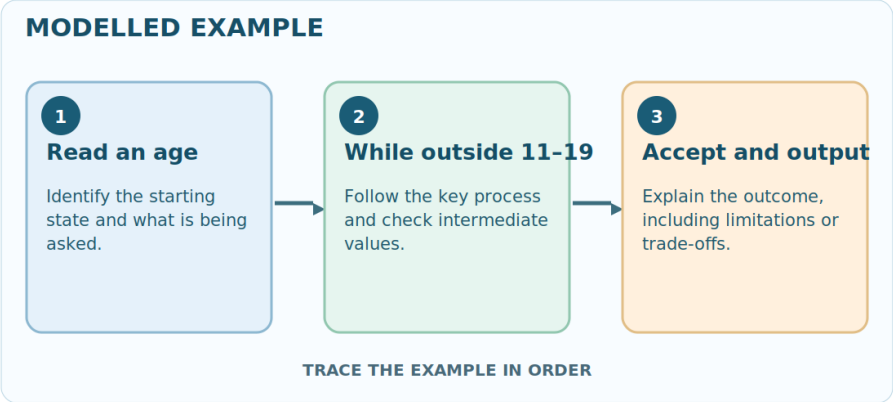
Problem

Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.

Follow the worked method

1. Model answer — age = int(input("Age")) — while age < 11 OR age > 19 — endwhile — print(age) — Note: this handles numeric range, not a non-numeric input error without extra handling. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write an algorithm accepting a positive integer (at least 1) and outputting it. [6]

Question 2

Explain why 'WHILE age < 11 AND age > 19' fails as range validation. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Extend a validation algorithm to reject non-integer input, describe robust behaviour and supply three tests. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

When rejecting values outside a range, the two invalid inequalities should usually be joined with AND.

Use key vocabulary from memory

1. Explain validation in your own words.

2. Explain robustness in your own words.

3. Explain iteration in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.